

Massive Upper GI Bleed



Section I: Scenario Demographics

Scenario Title:	Massive Upper GI Bleed		
Date of Development:	10/03/2015		
Target Learning Group:	☐ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☒ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Cheryl French
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Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To manage a patient who presents to the ED with a massive UGIB.
CRM Objectives:	1) Anticipate and appropriately plan for intubation in a patient with a significant UGIB 2) Consider a massive transfusion strategy in a non-trauma context
Medical Objectives:	1) Employ adjunctive medications in the treatment of an undifferentiated massive UGIB including octreotide/somatostatin analogues, intravenous proton pump inhibitor, and broad-spectrum antibiotics. 2) Prioritize airway management and Blakemore tube insertion for the stabilization of a patient with a massive UGIB prior to definitive therapy. 3) Recognize the importance of limited crystalloid therapy and early blood transfusion in the setting of hypovolemic shock secondary to UGIB.

Case Summary: Brief Summary of Case Progression and Major Events

A 58-year-old male known for alcoholism presents to the emergency department with a two-day history of hematemesis with an active, massive upper GI bleed due to esophageal varices. The patient deteriorates into hypovolemic shock requiring medical management, blood transfusions, intubation for airway protection and insertion of a Blakemore tube before definitive management.

References

- 1) Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- 2) Chavez-Tapia NC, Barrientos-Gutierrez T, Tellez-Avila FI, Soares-Weiser K, Uribe M. Antibiotic prophylaxis for cirrhotic patients with upper gastrointestinal bleeding. *Cochrane Database of Systematic Reviews* 2010, Issue 9.
- 3) Chase, C. (2014, September 2). Management and Dispo of Upper GI Bleed - emdocs. Retrieved May 20, 2015, from <http://www.emdocs.net/management-dispo-upper-gi-bleed/>



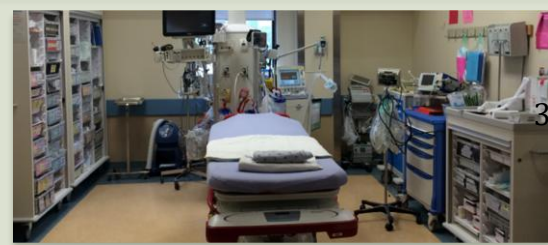
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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case			
You are working in a tertiary care centre emergency department with full consultant services. A patient is brought into the ED with active hematemesis. He is triaged to the resuscitation area.			
B. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin ☒ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☒ Physical ☐ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
Nurse	Simulates copious hematemesis with kidney basin full of blood and prompts the team to recognize ongoing massive hematemesis.		
C. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
D. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☒ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☒ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☒ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ ET Tubes	☒ Sengstaken-Blakemore tube	
☐ Intraosseous Set-up	☐ LMA	☐ Other:	
E. Moulage			
Hematemesis simulated by fake blood in kidney basin. If mannequin allows, continuous flow of red liquid from oral ports.			
F. Approximate Timing			
Set-Up:	5 min	Scenario:	15 min
		Debriefing:	12 min

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Section V: Patient Data and Baseline State

A. Patient Profile and History			
Patient Name: Franklin Jones		Age: 58	Weight: 80kg
Gender: ☒ M ☐ F		Code Status: Full	
Chief Complaint: Massive Hematemesis			
History of Presenting Illness: The patient presents with a 2-day history of vomiting which began as coffee grounds and progressed to bright red. His last drink was 3 hours ago and he stopped because of vomiting. He complains of general chest discomfort and nausea. He insists that he is "fine" and just "needs a beer".			
Past Medical History:	Alcoholism (10-20 beers/day)	Medications:	Hydrochlorothiazide 25mg QD.
	Hypertension		
Allergies: None			
Social History: Alcohol abuse, 1PPD cigarette smoking X 40 years, no Hx of recent illicit drug use			
Family History: Non-contributory			
Review of Systems:	CNS:	Alert, but confused and seems intoxicated	
	HEENT:	Nil	
	CVS:	Central general chest discomfort.	
	RESP:	Nil	
	GI:	Hematemesis for 2 days, coffee grounds → bright red blood, ongoing.	
	GU:	Nil	
	MSK:	Nil	INT:
B. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
		☐ Monitor on Standard Display	
HR: 115/min	BP: 105/60	RR: 24/min	O ₂ SAT: 96%
Rhythm: Sinus	T: 37.8°C	Glucose: 10 mmol/L	GCS: 13 (E3 V4 M6)
General Status: Alert but drowsy, mild confusion, smells of EtOH and clinically intoxicated			
CNS:	GCS 13, Drowsy, no FND.		
HEENT:	Presence of blood in oropharynx and nares.		
CVS:	Normal S1 and S2, no EHS		
RESP:	GAEB, no adventitia		
ABDO:	Soft, non-tender, +caput medusa, cirrhotic habitus		
GU:	Normal		
MSK:	Normal	SKIN:	Jaundiced, cool skin.

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: Sinus Tach HR: 115/min BP: 105/60 RR: 24/min O ₂ SAT: 96 % T: 37.8°C	Alert, drowsy, intoxicated, GCS: 13 Active intermittent Bright red blood hematemesis	Learner Actions ☐ Monitors/Full vital signs ☐ Physical exam ☐ 2 large bore IV access ☐ IV NS Fluid bolus ☐ CXR/AXR ☐ EKG ☐ Blood work ordered, including type & screen + Coags	Modifiers <i>Changes to patient condition based on learner action</i> Triggers <i>For progression to next state</i> - 2 minutes → 2. Hemorrhagic Shock
2. Hemorrhagic Shock HR → 140 over 30 seconds BP → 85/50	1 episode of copious hematemesis GCS: 7 (E1V3M4)	Learner Actions ☐ IV NS Fluid bolus #2 ☐ Blood Transfusion 2U ☐ Massive Transfusion protocol activation ☐ IV PPI (Bolus + infusion) ☐ Central line access ☐ Octreotide bolus + infusion ☐ Intubation ☐ ±Vasopressin infusion	Modifiers - Blood transfusion/MTP initiation → HR 120, BP 95/60 - IV NS bolus w/o blood → HR 125, BP 90/55 transiently. Triggers - Intubation → 3. Intubation - 5 minutes w/o intubation → 6. PEA arrest
3. Intubation <u>Display</u> (if possible) V _t : 500ml RR → 12 FiO ₂ : 1.0 EtCO ₂ = 50, normal waveform	Pt requires significant suction, able to intubate via direct laryngoscopy	Learner Actions ☐ 2 suction tips prepared ☐ Difficult Airway Kit ☐ Appropriate induction med ☐ Appropriate paralytic med ☐ Sengstaken-Blakemore tube ☐ GI/ICU/Surgery consult	Modifiers Triggers - Sengstaken-Blakemore tube → Sengstaken-Blakemore tube - 5 minutes without Blakemore tube placement after intubation → 6. PEA Arrest
4. Sengstaken-Blakemore tube		Learner Actions ☐ Preparation (HOB at 45°, test balloons, check tube markings) ☐ Placement of tube ☐ CXR ☐ IV Antibiotics (Ceftriaxone)	Modifiers Triggers - Blakemore tube confirmed with CXR → 5. Resolution.
5. Resolution HR → 110 BP → 105/80	Pt transferred for emergent endoscopy.	☐ ±PCC/Vit K ☐ ±TXA	END SCENARIO
6. PEA Arrest HR → 130 BP → 0/0 O ₂ SAT → 0	Pt has no pulse	Learner Actions ☐ CPR ☐ Epinephrine	END SCENARIO PRN GI/ICU arrive and declare patient unsalvageable



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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 129	K: 4.5	Cl: 104	HCO ₃ : 23	BUN: 40	Cr: 200	Glu: 10
Ca:		Mg:		PO ₄ :		Albumin:
VBG	pH: 7.21		PCO ₂ : 32	PO ₂ : 40	HCO ₃ : 23	Lactate: 4.0
WBC: 12.1		Hg: 50		Hct: 22%		Plt: 220
INR: 1.8		PTT: 51				
AST: 175	ALT: 73		Bili (T): 756	ASA: 0	APAP: 0	EtOH: 40
Images (ECGs, CXRs, etc.)						
EKG – Sinus Tachycardia http://cdn.lifeinthefastlane.com/wp-content/uploads/2011/12/sinus-tachycardia.jpg				CXR – Prior to intubation https://emsimcases.files.wordpress.com/2015/04/normal-cxr-male.jpg		
Post Intubation X-ray https://emsimcases.files.wordpress.com/2015/03/post-intubation-male.png				Blakemore Tube X-ray http://images.radiopaedia.org/images/585576/652c659aa92ffac9625a44acfa7b9a_big_gallery.jpg		

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To manage a patient who presents to the ED with a massive UGIB.
CRM Objectives:	1) Anticipate and appropriately plan for intubation in a patient with a significant UGIB 2) Apply a massive transfusion strategy in a non-trauma context
Medical Objectives:	1) Employ adjunctive medications in the treatment of a likely variceal UGIB including octreotide/somatostatin analogues and broad-spectrum antibiotics. 2) Prioritize airway management and Blakemore tube insertion for the stabilization of a patient with a massive UGIB prior to definitive therapy. 3) Recognize the importance of limited crystalloid therapy and early blood transfusion in the setting of hypovolemic shock secondary to UGIB.
Sample Questions for Debriefing	
1) What are the medical management options for a significant variceal bleeding? 2) What are the evidence based outcomes for the use of octreotide in UGIB? Antibiotics? PPI? 3) What is a massive transfusion protocol? When and how would you initiate an MTP for a non-trauma patient? 4) What are the airway considerations in a patient with a massive UGIB? 5) Describe the set up and insertion of a Sengstaken-Blakemore tube. 6) Describe the coagulopathy associated with cirrhosis and its effect on management of the cirrhotic UGIB.	
Key Moments	
- Hemorrhagic Shock	
- Intubation	
- Sengstaken-Blakemore tube insertion	

